ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

# MECHATRONICS

## PAPER 1 · PRACTICE D · MARKING GUIDELINE

**MARKS: 100**

**Marking guideline for the Academic Central NC(V) Level 2 Mechatronics paper 1 · practice d.**

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training marking guideline.

**How to mark:** each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers, sensible design choices and neat, recognisable diagrams.

## SECTION A

## QUESTION 1

**1.1** 1.1.1 A ✓ 1.1.2 B ✓ 1.1.3 A ✓ 1.1.4 C ✓ 1.1.5 A ✓ 1.1.6 C ✓ 1.1.7 D ✓ 1.1.8 B ✓ 1.1.9 C ✓ 1.1.10 B ✓ (10)

**1.2** 1.2.1 True ✓ 1.2.2 False ✓ 1.2.3 True ✓ 1.2.4 True ✓ 1.2.5 False ✓ 1.2.6 True ✓ 1.2.7 True ✓ 1.2.8 True ✓ 1.2.9 False ✓ 1.2.10 False ✓ (10)

**1.3** 1.3.1 C ✓ 1.3.2 G ✓ 1.3.3 B ✓ 1.3.4 I ✓ 1.3.5 E ✓ 1.3.6 H ✓ 1.3.7 J ✓ 1.3.8 A ✓ 1.3.9 D ✓ 1.3.10 F ✓ (10)

**TOTAL SECTION A: [30]**

## SECTION B

## QUESTION 2

**2.1** In order ✓✓✓✓✓: identify the problem (design brief); research / investigate; develop ideas and choose the best solution; plan and make a prototype; test and evaluate (and improve). (5)

**2.2** Any FOUR ✓✓✓✓: moves a 300 kg gate 4,5 m; opens in less than about 25 s; works by remote control; stops at both ends; stops / reverses when the beam is broken; runs on battery during load-shedding; weatherproof; manual release. (4)

**2.3** Inputs ✓✓: remote receiver, open and closed limit switches, safety beam (and push button). Controller ✓. Outputs ✓✓: relays / motor driver to the DC motor, warning light. Power supply / battery and correct arrows ✓ (6)

**2.4** To warn people and drivers that the gate is moving so that nobody is hit or trapped ✓ (1)

**2.5** Stop the motor immediately ✓ and then reverse / open the gate again (or wait until the beam is clear) ✓. (2)

**[18]**

## QUESTION 3

**3.1** $T = F \times r$ ✓ $= 400 \times 0{,}03$ ✓ $= 12\ N{\cdot}m$ ✓ (3)

**3.2** $N = 1\,400 \div 20$ ✓ $= 70\ r/min$ ✓ (2)

**3.3** $v = \pi dN \div 60$ ✓ $= \pi \times 0{,}06 \times 70 \div 60$ ✓ $= 0{,}22\ m/s$ ✓ (3)

**3.4** $t = s \div v$ ✓ $= 4{,}5 \div 0{,}22 = 20{,}47\ s$ ✓ (accept 20,45 s with 0,22) (2)

**3.5** $P = F \times v \checkmark = 400 \times 0{,}22 \checkmark = 87{,}96\ W\ (\approx 88\ W) \checkmark$ (3)

**3.6** $T_{motor} = 12 \div 20 \checkmark = 0{,}6\ N{\cdot}m \checkmark$ (2)

**3.7** The motor turns too fast (1 400 r/min) for the gate ✓; the gearbox lowers the speed to a safe gate speed ✓ and increases the torque so that a small motor can move the heavy gate ✓. (3)

**[18]**

## QUESTION 4

**4.1** Controller output (through a transistor) to the relay coil ✓; flywheel diode across the coil ✓; 12 V supply ✓; fuse in series ✓; relay contacts in series with the motor ✓; correct symbols, neat, labelled ✓ (6)

**4.2** Reverse the polarity of the supply to the motor ✓ using two relays (one for each direction) or a DPDT relay / H-bridge ✓, with an interlock so that both directions can never be on together ✓. (3)

**4.3** A limit switch is fitted at each end of the track ✓; when the gate reaches it, the switch changes state ✓ and the controller (or the switch in the circuit) switches off the motor for that direction ✓. (3)

**4.4** $\text{Charge per cycle} = 8 \times (20 + 20) = 320\ As \checkmark = 320 \div 3\,600 = 0{,}089\ Ah \checkmark;$ $\text{cycles} = 7 \div 0{,}089 \checkmark = 78\ \text{complete cycles} \checkmark$ (4)

**4.5** 10 A ✓ – it carries the normal 8 A without blowing but blows quickly on a fault (5 A would blow in normal use; 30 A would not protect the motor) ✓ (2)

**[18]**

## QUESTION 5

**5.1** Any FOUR ✓✓✓✓: infrared safety beam; obstacle / current sensing that stops or reverses the gate; flashing warning light; manual release; limit switches; anti-crush rubber edge; overload protection; secure remote codes. (4)

**5.2** (a) galvanised steel (or aluminium) ✓ – strong and resists rust outdoors ✓ (b) nylon or steel rack ✓ – nylon is quiet and does not rust; steel is very strong for heavy gates ✓ (4)

**5.3** Any FOUR ✓✓✓✓: clean the track and remove stones; check and clean the safety-beam sensors; check rack and pinion for wear and alignment; test the back-up battery; test the limit switches and manual release; tighten bolts; lubricate moving parts as recommended. (4)

**5.4** Any FOUR ✓✓✓✓: obstruction on the track; safety beam blocked or dirty; flat battery / power failure; overload tripped because the gate is stiff; faulty or loose wiring; pinion slipping off the rack; a limit switch faulty or out of position. (4)

**[16]**

**TOTAL SECTION B: [70]**

**TOTAL: 100**

**END OF MARKING GUIDELINE**